Why Does DPO Make Olmo 3 Overconfident?

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Summary

Olmo 3 7B's Instruct DPO stage is the largest single source of the model's overconfidence on multiple-choice questions. Most of the preference pairs it trains on are ones the SFT model already gets right, meaning it already gives the chosen response a higher probability than the rejected one, yet DPO gives them the same starting weight as the pairs it gets wrong. We interpret this as DPO reinforcing preferences the model already holds, which also makes it more confident where it commits to an answer. Down-weighting the pairs SFT already gets right removes this overconfidence when trained on 25% of Ai2's DPO pairs, with no detectable loss in what the model learns (one run). Trained on all of them, it removes about 60% of it on MMLU, and responses get much longer.

- **Where.** Instruct DPO is the largest single calibration step in the pipeline, about half of the post-training rise in calibration error on MMLU. The temperature needed to recalibrate the model (T*, where 1 = calibrated) rises from 1.52 to 2.34, and calibration error (ECE) from .115 to .205. The chosen letters, accuracy and the ranking of right vs wrong answers barely change; mostly confidence rises. It happens at the token where the model commits to an answer (a multiple-choice letter, the first word of a short answer), not on ordinary text, and, on MMLU, is much smaller when the model reasons first.
- **Why.** DPO measures progress relative to the SFT model, so pairs SFT already gets right start with the same weight as the ones it gets wrong. In a pre-registered three-seed test on training sets matched for data source and length, sets in which 99% vs 18% of pairs are already right give T* 2.26 vs 1.46; five of six predictions held. What matters is whether SFT already gets a pair right, not by how much, and the reference is what makes DPO keep pushing on those pairs. SimPO, which has none, stays at SFT's calibration, and adding the SFT reference to it brings back all of plain DPO's overconfidence.
- **Intervention.** If the account is right, training less on already-right pairs should prevent the overconfidence. Down-weighting each pair by how strongly SFT already prefers its chosen response keeps T* at SFT's level when trained on 25% of Ai2's DPO pairs (1.50 vs 1.85 for plain DPO), with no detectable loss in learning, accuracy, IFEval or judged

quality. Trained on all ~260K pairs, it removes about 60% of the MMLU rise, but responses grow to 2.4× SFT's length, IFEval is 3.5 points below DPO, and an LLM judge prefers SFT's answers (40% win rate, 50% after adjusting for length). On BigBenchHard, our only capability check at that scale, it scores higher than DPO, but it reasons 4.4× longer to get there. So we treat it as a test of this explanation; it is not ready as a recipe change.

- **Inside the model.** In the released DPO model, the scores for the four answer letters are about 1.44× SFT's, with the same winning letter. Among attention heads, the change sits in the ones SFT already uses to pick its answer. If we run SFT but give its 30 most answer-relevant heads the outputs they produce in the DPO model, 35% of DPO's extra confidence appears, vs 1% for 30 random heads. The MLP part is spread out, since any 12 MLP layers carry 28–53% of it. Simply turning those heads' outputs up does not reproduce the change.

All experiments use Ai2's released Olmo 3 checkpoints and data (Team Olmo, 2025); our DPO reruns use Ai2's open-instruct code and recipe. Prior work has shown that DPO hurts calibration (Xiao et al., 2025) and proposed new losses to counter it; what is new here is the located cause (which pairs, and that measuring progress against the SFT reference is what matters) and an intervention that tests it by changing only how pairs are weighted (see Related work).

Key terms

Term	Meaning
Preference pair ("pair")	A prompt with a chosen (preferred) and a rejected response; DPO's training data
Preference margin ("margin")	A model's average per-token log-probability of the chosen response minus that of the rejected one. Positive = the model prefers the chosen response
Agreed / disagreed pair	A pair whose preference margin under the SFT model, before DPO, is positive / negative (also: pairs SFT already gets right / gets wrong). About 63% of Olmo 3's Instruct DPO pairs are agreed
Letter scale	How much a model's A–D letter scores (each minus their mean) are multiplied relative to SFT's. Above 1 = more confident, same winning letter

Term	Meaning
Letter gap	The predicted letter's score minus the mean of the other three
Top-two gap	The difference between the two highest next-token log-probabilities where the model answers; used for answers that are not letters
Answer token	The token where the model commits to its answer: a multiple-choice letter, or the first token of a short answer
Learned	Of 1,000 held-out pairs, 372 are disagreed (SFT gets them wrong). Learned = the share of those a trained model now gets right. It is not a capability score: it checks that a run actually learned the preferences, so that lower overconfidence cannot simply mean less training. It favors methods that up-weight disagreed pairs, since those are the pairs it counts; agreed held-out pairs stay about 97–99% right in every run
Margin slope	Fit each held-out pair's margin after training against its margin under SFT. Above 1 = existing margins were multiplied; about 1 = shifted
Small test	A DPO run on about 8,000 pairs at 2–8× Ai2's learning rate, stated with each result (64 steps, about an hour on one GPU). It reproduces the released DPO model's pattern: T* up, accuracy and AUROC flat
25% scale	A rerun on 65,000 of Ai2's ~260K Instruct DPO pairs with Ai2's hyperparameters (LR 1e-6, 508 steps)

Setup

Olmo 3 is the right testbed because every stage is public: pretraining, midtraining, base, SFT, DPO and RL checkpoints, plus the exact SFT and DPO data (Team Olmo, 2025). That allows both localizing a change to one stage and rerunning that stage with modified data

or loss.

Models. Olmo 3 7B pipeline checkpoints: pretrain → midtrain → base (the released checkpoint after midtraining) → Think-SFT → Instruct-SFT (trained on top of Think-SFT) → Instruct-DPO → Instruct (after RL). The Think branch (models that reason at length before answering) has its own DPO and RL stages; Instruct models answer directly.

Calibration metrics on 5-shot MMLU (Hendrycks et al., 2021) and ARC-Challenge (Clark et al., 2018), scoring the A–D letter probabilities:

Metric	What it measures	Calibrated value
ECE (15 bins; Naeini et al., 2015)	Gap between confidence and accuracy	0
T*	One temperature that best calibrates the model, fit on half the questions and checked on the other half	1 (above 1 = overconfident)
AUROC	Whether confidence ranks right above wrong answers; unaffected by temperature	higher is better

T* is a measurement, not a proposed fix (see Limits). It says how much sharper than calibrated the answer probabilities are. One temperature (temperature scaling; Guo et al., 2017) removes most of the calibration error of post-trained checkpoints (66–85% on MMLU 5-shot; 79% for Instruct-DPO), so their miscalibration is mostly uniform sharpening, which T* captures in one number less sensitive to binning than ECE (Nixon et al., 2019; Roelofs et al., 2022); we report both. Three other formats check that results are not an artifact of 5-shot letters: chat-format questions (the letter is read right after an "Answer:" prefix, with no reasoning), cloze scoring (scoring each option's text instead of its letter, averaged per token) and self-consistency (how often 8 sampled reasoning chains agree; Wang et al., 2023).

Our DPO runs use Ai2's open-instruct code (Lambert et al., 2024; commit 11826255) and Ai2's Instruct DPO hyperparameters: DPO (Rafailov et al., 2023) on per-token-averaged log-probabilities, β = 5, batch 128, one epoch, 10% warmup. Small tests and 25%-scale runs are defined in Key terms. Small tests are scored on a fixed 3,000-question MMLU subset, where SFT's T* is 1.53 (1.52 on all 14,042 questions); each run is compared with SFT

on the same set. Repeating a run on the same data changes T* by about ±0.02; small tests on different random subsets of pairs differ by about 0.2, so we compare runs on the same pairs wherever possible.

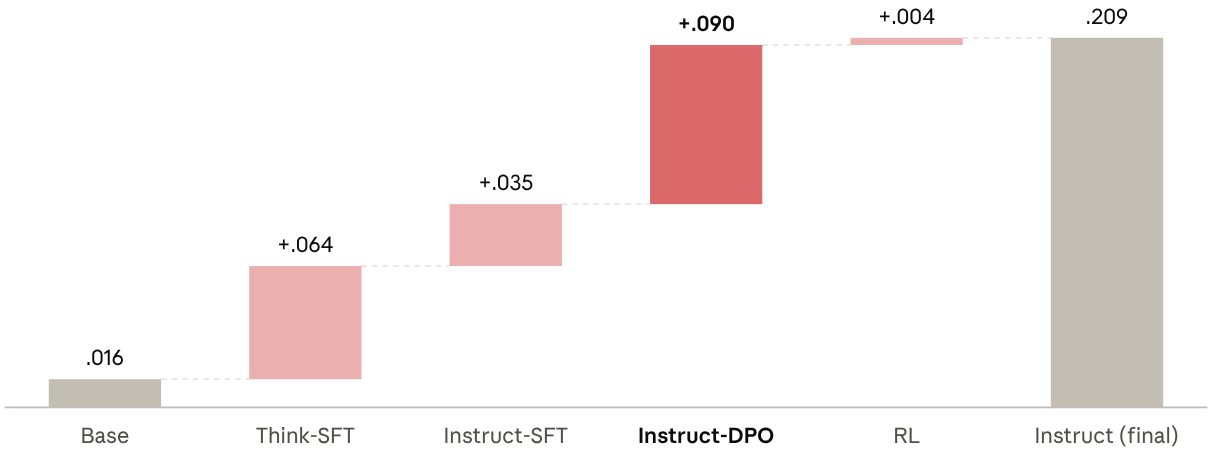
Where: DPO sharpens the model's answers

Instruct DPO is the largest single calibration step in the pipeline, and it mostly sharpens: the chosen answers stay the same and the confidence goes up.

Instruct DPO adds the most miscalibration of any stage

MMLU 5-shot ECE, base to final Instruct: .016 to .209; the DPO step is 47% of the rise

Grey = ECE at a checkpoint; red = what a stage adds



Pipeline order, Instruct branch (Instruct-SFT is trained on top of Think-SFT)

Full MMLU 5-shot (14,042 questions), A–D letter probabilities, released Olmo 3 7B checkpoints

- **Mostly sharpening.** From Instruct-SFT to Instruct-DPO, T* rises 1.52 → 2.34 while accuracy and AUROC stay flat (AUROC .77 in both). Confidently wrong answers rise too (5.1% → 14.6% of chat-format answers at confidence ≥ 0.9), as sharpening implies. SFT is different. From base to Instruct-SFT, it both sharpens and hurts ranking (AUROC .80 → .77).
- **Branch and stage.** Instruct DPO is the largest single step on MMLU 5-shot and on ARC; on MMLU 0-shot the Think-SFT step is about as large (Think-SFT +.111 vs Instruct DPO +.108 ECE). The Think branch's own DPO stage adds only 20–40% as much ECE as Instruct DPO, and RL adds about nothing on either branch.
- **At the answer, not in ordinary text.** Measured the same way, as the gap between the model's top two choices at one token, DPO widens it ×1.51 at a multiple-choice letter and ×1.44 at the first token of a short free-form answer (prompts from DPO's training

data), but only ×1.08 on ordinary text (wikitext). Scoring each option's whole text instead of its letter (cloze) also shows little sharpening: T* ×1.10, vs ×1.54 for letters.

- **Smaller after reasoning, on MMLU.** On MMLU, DPO adds +0.114 to SFT's excess confidence (mean confidence minus accuracy) for answers given immediately, but only +0.021 for answers given after reasoning, where confidence is the agreement among 8 sampled reasoning chains. The two confidence measures differ. Measuring immediate answers the same way (agreement among 8 samples at the same temperature, 0.7) gives +0.072 (95% interval +0.060 to +0.086), while after reasoning the difference is within noise of zero (+0.021, interval −0.005 to +0.046; −0.014 if chains that state no answer count as votes). On ARC the two are about the same (+0.031 vs +0.028), so the drop is not general.
- **Reproduced.** Our own 25%-scale rerun of the DPO stage shows the same pattern: T* 1.52 → 1.85, with accuracy, AUROC and IFEval (Zhou et al., 2023) unchanged.

Why: DPO keeps training on pairs SFT already gets right

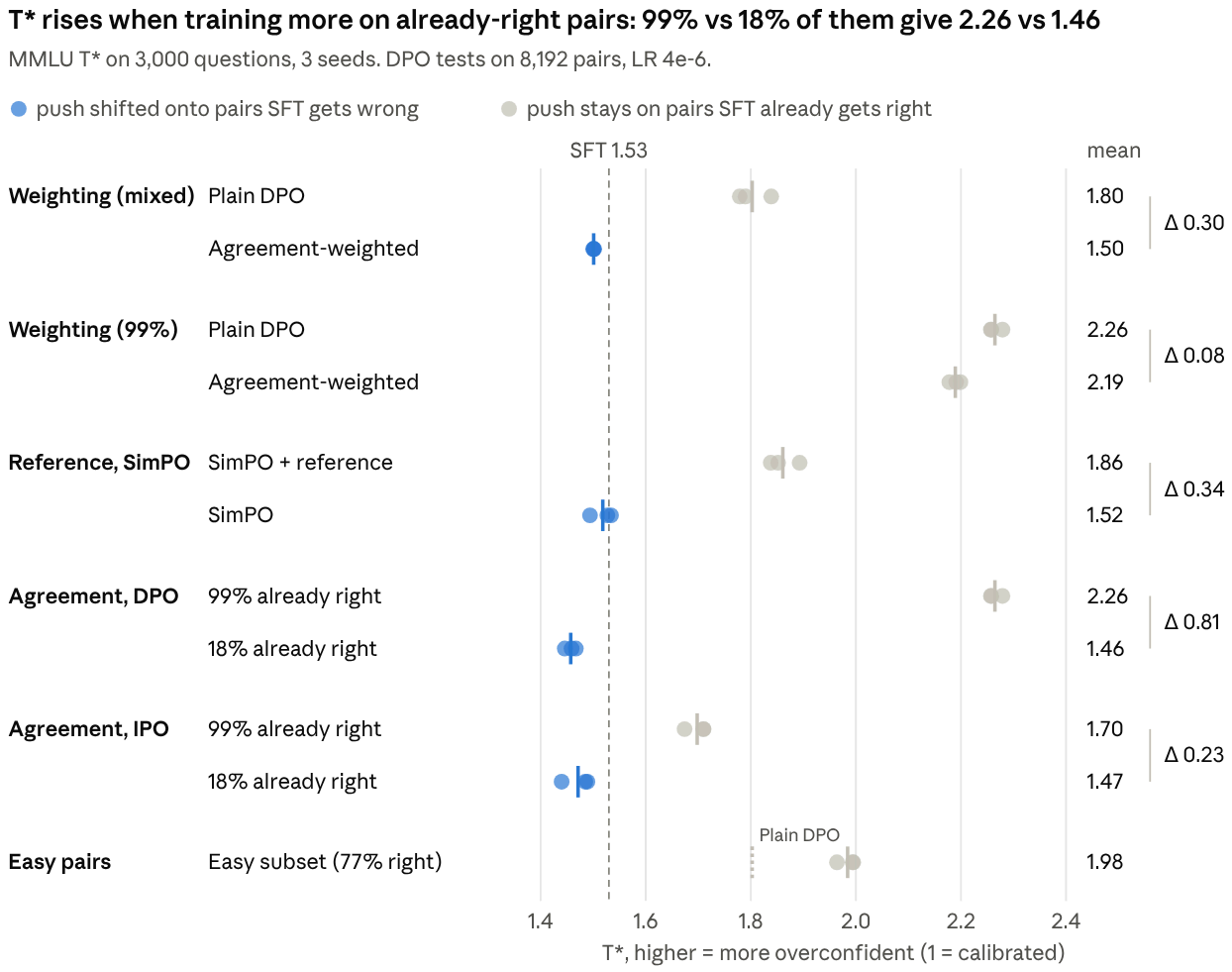
Our account, which the rest of this write-up tests, is that DPO keeps widening margins on pairs the SFT model already gets right (agreed pairs), and that this is what raises its confidence. DPO compares the model being trained, π, with a frozen reference model π_ref (here the SFT model), using the log-probabilities of the chosen response y_c and the rejected response y_r (averaged per token in Olmo 3's recipe):

```
\mathcal{L} = -\log\sigma\Big(\beta\big[\log\tfrac{\pi(y_c)}
{\pi_{\mathrm{ref}}(y_c)} - \log\tfrac{\pi(y_r)}{\pi_{\mathrm{ref}}
(y_r)}\big]\Big)
```

At the start of training π equals π_ref, so the bracket is 0 and every pair gets the same gradient weight, σ(0) = 0.5, whether or not SFT already prefers the chosen response. Nothing in the loss tells DPO that an agreed pair is already right, so it keeps widening those margins. We interpret the results that follow this way: when most pairs are agreed, the cheapest way to widen many margins at once is to strengthen the components that already produce the model's preferences, and that also raises confidence on unrelated multiple-choice answers. Disagreed pairs instead need the model to change which response it prefers. The mechanism section shows this strengthening is not a simple scale-up.

Pre-registered replication

Before training, we wrote down six predictions with thresholds and a stop rule (file hash recorded), then ran 27 small tests at LR 4e-6 (4× Ai2's rate), three seeds each. Besides plain DPO, they use three variants: agreement-weighted DPO, which multiplies each pair's loss by σ(−5 × its SFT margin) (details in Intervention); SimPO (Meng et al., 2024), which has no reference model and lets a pair's gradient fade once the model's own margin clears a target; and IPO (Azar et al., 2024), a squared loss that stops pushing a margin once it reaches a target. Five of the six predictions held, and two follow-up predictions, each written before its own runs, did not (the last two rows); seed-to-seed spread in T* was 0.06 or less.



27 small DPO tests from Instruct-SFT (8,192 pairs, LR 4e-6, seeds 1–3); T* on the 3,000-question MMLU subset; why_tests/WHY_RESULTS.md

Test (3 seeds each)	Predicted	Outcome
99% vs 18% already-right pairs, matched on data source and length	T* gap ≥ 0.3, every seed	Confirmed; the 99% set also learns more (22.9% vs 19.3%)
Agreement-weighted vs plain DPO	≥ 0.15 lower T*, ≤ 2 points less learning	Confirmed; learns 1.4 points less
SimPO (no reference) vs plain DPO	≥ 0.1 lower T*, ≤ 3 points less learning	Confirmed; learns 2.0 points more
SimPO with DPO's reference added	restores ≥ half of plain's excess	Confirmed: all of it (121% of plain's excess)
IPO (caps each margin) on the same two sets	below DPO; gap still ≥ 0.15	Confirmed
"Easy" pairs (high margin under a DPO-trained model)	more overconfident with no increase in learning	Not supported: learns 29.2% vs 17.7%
Follow-up: agreement weighting on the 99%-already-right set	≥ 0.2 lower T* than plain DPO on the same set	Not supported: 2.26 → 2.19
Follow-up: the same on a 100%-already-right set (one seed)	≥ 0.2 lower T* than plain DPO on the same set	Not supported: 2.32 → 2.24

The SimPO pair is the sharpest test: the two runs differ only in whether each pair's margin is measured relative to the SFT model, and that change alone moves T* from SFT's level to slightly above plain DPO's. IPO removes about 70% of the agreement effect, not all of it, and learns about half as much as DPO (12–14% vs 19–23%). The easy-pairs prediction failed on its premise: those pairs taught much more, at less overconfidence per unit of learning than plain DPO. At this learning rate agreement weighting and SimPO do not lengthen responses (400–402 tokens vs plain's 403).

The two follow-ups show that what matters is whether SFT already gets a pair right, not by how much. On almost or fully agreed sets, agreement weighting can only move weight from strongly to weakly agreed pairs, and T* barely moves; on mixed data, where it moves weight onto disagreed pairs, T* falls to SFT's level. Removing the 73 disagreed pairs from the 99% set leaves the weighted result about the same (within 0.1, one seed), so those few pairs were not what weighting acted on there.

Because agreed pairs get the same weight as disagreed ones, most of plain DPO′s margin growth goes to pairs SFT already gets right. In our 25%-scale rerun, the average margin grows +0.34 per token on agreed held-out pairs but only +0.13 on disagreed ones.

The earlier single-run tests that first pointed to agreement (agreement splits, label swaps and how held-out margins grow), and comparisons with Olmo 3′s Think DPO and delta-learning pairs, are in Appendix A.

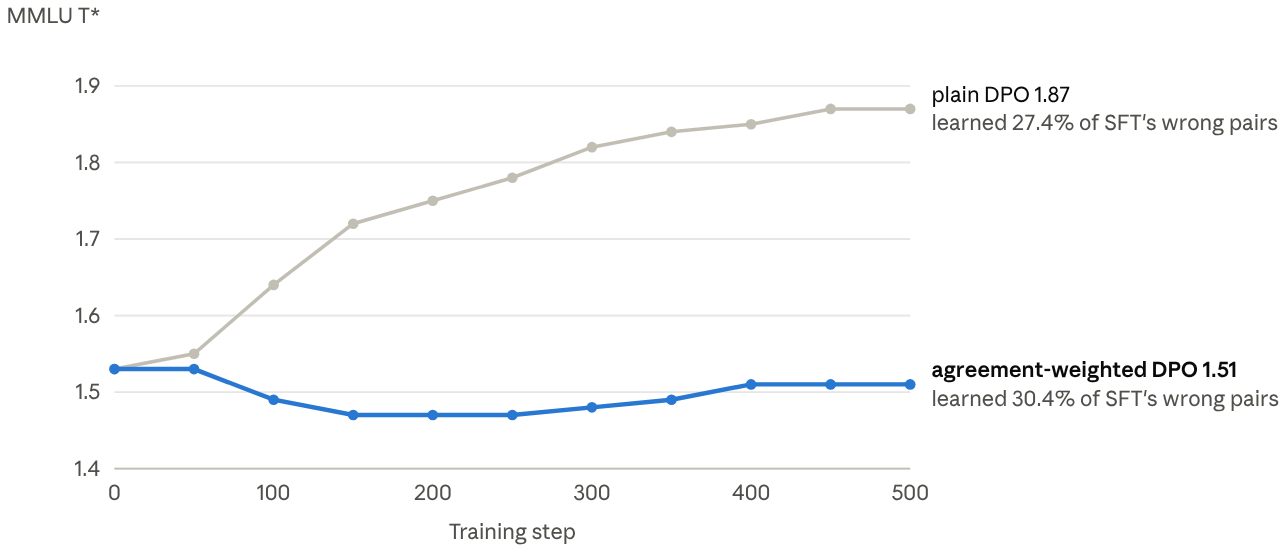
Intervention: train less on pairs SFT already gets right

If the account is right, a DPO variant that trains less on agreed pairs should remove the overconfidence without reducing learning. Agreement-weighted DPO multiplies each pair′s loss by σ(−5 × its SFT margin), with weights rescaled to average 1. Before rescaling, a pair that SFT already strongly prefers (margin +0.5) gets weight 0.08 and a disagreed pair (margin −0.3) gets 0.82, so agreed pairs drop from 64% to 31% of the total weight and disagreed pairs get about 1.9× their plain-DPO weight; nothing else changes. The 5 is a separate constant from DPO′s β = 5, picked once and not tuned. In small tests it keeps SFT′s calibration (see the replication in Why); here we test it with Ai2′s recipe at 25% and full scale. It works by moving weight onto disagreed pairs, so on data with almost none it does little (see Why).

The agreement-weighted version then held at 25% scale, with Ai2′s exact recipe and data order:

At 25% scale, weighted DPO stays at SFT calibration while plain DPO climbs

65K pairs, Ai2 recipe (LR 1e-6, 508 steps), same data order; MMLU T* at each 50-step snapshot



Plain vs agreement-weighted DPO at 25% scale. T* at each snapshot is measured on the 3,000-question MMLU subset (SFT 1.53), so values differ slightly from the full-MMLU table below; learned share at the final model

Final model (full evals)	SFT	Plain DPO	Agreement-weighted
MMLU T* / ECE	1.52 / .115	1.85 / .155	**1.50 / .108**
ARC T*	1.17	1.42	**1.14**
Chat-format MMLU T*	1.90	2.70	**1.84**
MMLU accuracy / AUROC	.608 / .768	.613 / .768	.613 / .768
IFEval (prompt, strict)	.769	.773	.782
Learned (SFT's wrong held-out pairs now right)	0%	27.4%	30.4%
LLM-judge win rate vs SFT (300 held-out prompts, length-adjusted)	—	.57	.59
Mean response length (tokens)	371	426	448

Mean responses are 5% longer than plain DPO's. The learning difference (30.4% vs 27.4%) is within noise, so the claim at this scale is no detectable loss of learning, from one run of each.

At full scale it holds only partly

We then ran agreement-weighted DPO on all 259,922 Instruct pairs with Ai2's recipe (2,031 steps). The comparison is Ai2's released DPO model, which is plain DPO on the same data; unlike at 25% scale, it is not our own rerun, so differences in seed, data order and hardware are not controlled. Held-out pairs cannot measure learning here, since the full data contains them, so BigBenchHard stands in for capability.

Final model (full evals)	SFT	Released DPO	Agreement-weighted
MMLU T* / ECE	1.52 / .115	2.34 / .205	**1.80 / .150**
ARC T*	1.17	1.87	1.53
Chat-format MMLU T*	1.90	3.77	**1.87**
MMLU accuracy / AUROC	.608 / .768	.615 / .766	.609 / .767
IFEval (prompt, strict)	.769	.789	.754
BigBenchHard, Ai2's protocol (6,511 questions)	.448	.567	**.641**
Mean response length, chat (tokens)	371	630	907
Mean response length, BigBenchHard (tokens)	215	607	2,685
LLM-judge win rate vs SFT, 300 prompts (raw / length-adjusted)	—	.55 / .55	.40 / .50

The weighting removes about 60% of DPO's rise in overconfidence on MMLU (in log T* and in ECE), about 40% on ARC, and all of it in chat format. Unlike the 25%-scale run, T* climbed during training, to about 1.85 by mid-run on the 3,000-question subset, and stayed there. IFEval is 3.5 points below DPO. An LLM judge (Zheng et al., 2023) comparing chat answers with SFT's on 300 prompts (training prompts for both full-scale models) prefers

SFT's: the weighted model wins 40%, released DPO 55%, and 50% vs 55% once length is adjusted for (a logistic fit on the log length ratio, similar in spirit to Dubois et al., 2024). Answers cut off at the token cap lose most, but finished ones also lose (46%). The largest cost is length, which we cannot yet explain: chat responses are 2.4× SFT's (26% hit our 2,048-token cap, so the true mean is higher), and BigBenchHard answers reason 4.4× longer than the released DPO model's. Most of those long answers do finish: 3.7% of BigBenchHard answers run past 16,384 tokens, and only 18 of those 243 are stuck repeating.

Appendix B describes the BigBenchHard setup and two explanations for the full-scale drift that the data rule out.

What this means. For the account: weighting removes most of the gain in DPO's answer signal at every scale we tested (about 80% at full scale), but at full scale most of DPO's normalization change survives (both terms are explained in Mechanism; the numbers are in Appendix C), and we have not tested whether that part depends on agreement. For the Olmo recipe: agreement weighting is cheap, since it reuses the SFT log-probabilities open-instruct already caches for the DPO loss and adds no extra model or forward pass, but at full scale it is not a drop-in change, since it makes responses much longer and lowers judged answer quality. For building preference data: how pairs are constructed sets how many are already right. Olmo 3's delta-learning pairs are 74% agreed, which is why training on them alone sharpens more.

Mechanism: what changes inside the model

In the released Instruct-DPO model, the scores (logits) for the four answer letters are 1.44 × SFT's after subtracting each question's mean over A–D (the letter scale; R^2 .96 on 1,000 MMLU questions). To see where this comes from, we split the letter gap into the direct contributions of every attention head and MLP (Elhage et al., 2021); we call the attention heads that contribute most to SFT's letter gap its answer heads. With the final normalization's scale held fixed, these contributions add up exactly to the gap; the change in that scale is treated separately below. A letter's score is the final residual's projection onto that letter's output direction, divided by the residual's norm, so DPO can raise confidence by writing more along the letter directions (signal) or by shrinking that norm (normalization: the parts of the residual unrelated to the answer shrink, so the answer part takes up more of it after normalization).

- **Direct contributions come from layers 15–31.** The letter gap grows from 3.01 to 4.46: +0.60 from attention, +0.86 from MLPs, almost nothing directly from layers 0–14. Earlier layers matter indirectly: inserting DPO's residual stream into SFT at layer L (activation patching; Heimersheim & Nanda, 2024), with SFT running the remaining layers, recovers 50% of DPO's extra letter scale at L = 18; the share rises roughly linearly from near zero at L = 9 to all of it at L = 31.
- **Among heads, the change is concentrated in SFT's own answer heads; among MLPs it is spread out.** Running SFT while giving selected components the outputs they produce in the DPO model on the same question, at the answer position (later layers still run as SFT), reproduces part of DPO's extra letter scale. SFT's top 30 answer heads (ranked by their contribution to SFT's letter gap; notation L22H6 = layer 22, head 6) give 35%, against 1% for 30 random heads. The late MLPs (layers 19–30) give 49%, but size-matched random sets of 12 MLP layers give 28–53% over three draws, so the MLP change is not concentrated in answer-specific layers. Top-30 heads plus late MLPs give 76% (30 random heads plus 12 random MLPs: 54%); all attention and MLPs in layers 15–31 give 100%.
- **Not a uniform scale-up.** Multiplying the same components' outputs in SFT by 1.3–1.5 barely raises confidence (top 30 heads: T* 1.60 → 1.65, vs DPO's 2.45 on the same questions). Scaling the late MLPs raises T* to 1.81 but damages ordinary text, and dividing these components in DPO barely lowers its T*. DPO changes what these components write, not just how strongly; what exactly changes is not yet characterized.

Appendix C splits DPO's change into signal and normalization, compares its size across models, and shows how agreement weighting changes it at 25% and full scale.

Limits and open questions

These results come from one model (Olmo 3 7B Instruct) and one preference dataset, and the intervention was tested on one 25%-scale run and one full-scale run, where it holds only partly. One 25%-scale run that removed very short, answer-first or degenerate rejected responses became more overconfident (T* 2.02 vs 1.85) for reasons the account does not explain.

- **Small runs, one dataset.** The causal tests use 8K-pair runs. Their main contrasts were repeated in a pre-registered test (three seeds, data matched for source and length, LR 4e-6, which is 4× Ai2's rate), but not at 25% or full scale, where the intervention has one run each.

- **Measuring whether DPO still did its job.** The learning metric is accuracy on held-out pairs SFT gets wrong, which comes from the same distribution as the DPO data and favors methods that up-weight disagreed pairs by construction. The LLM judge (gpt-6-luna, 300 prompts) only rules out large regressions: it rates our 25%-scale plain DPO run just 56% against SFT (57% after adjusting for length). The held-out pairs and judge prompts are held out from every small-test and 25%-scale run, but they are part of Ai2's full DPO data, so for the released DPO model and our full-scale run they measure training pairs; the full-scale comparison therefore relies on a standard benchmark instead (BigBenchHard). Apart from BigBenchHard, Ai2's standard post-training evals were not run.
- **The mechanism is partly open.** Transplants show where DPO's change lives and scaling shows it is not a uniform gain, but what those components now write differently is not yet characterized. The step from agreed preference pairs to multiple-choice confidence is an interpretation: on DPO's own training text, token-level predictions get slightly softer, not sharper.
- **Possible extra softening.** The agreement-weighted models are softer than SFT on ordinary text and at the first token of short answers (top-two gap ×0.96 and ×0.86 at 25% scale, ×0.91 and ×0.79 at full scale, where responses are also much longer). Part of their calibration may come from this general softening rather than only from removing DPO's gain, and the softening may be linked to the longer responses.
- **Why not just rescale the temperature afterward?** One post-hoc temperature fixes letter answers, but the released DPO model needs different values for different formats (T* 2.34 for 5-shot letters vs 3.77 in chat format), and generated text has no single temperature to fit.
- **SFT's share.** SFT also raises T* (1.03 → 1.52) and slightly hurts ranking; its cause is not established here.
- **Untested.** On-policy pairs, with both responses sampled from SFT and ranked by a judge (SFT is close to indifferent between its own samples, so about half would be agreed, and the account predicts less overconfidence than on Olmo 3's data), other model sizes, and calibration of long-form answers.

Related work and what is new

That RLHF-style post-training hurts calibration is well known (Kadavath et al., 2022; OpenAI, 2023). More recent work finds that base models are calibrated and that RL instruction tuning and chain-of-thought break this (Nakkiran et al., 2026), and that instruction tuning hurts calibration with the chat template adding to it (Sanz-Guerrero et

al., 2026), consistent with the higher T* we find in chat format. The closest prior work, Xiao et al. (2025), finds calibration degrading pretrained → SFT → DPO with DPO worst, attributes it to "preference collapse", and proposes a new calibration-aware fine-tuning loss; other "calibrated" DPO variants calibrate the scale of the implicit reward rather than confidence (e.g. Cal-DPO; Xiao et al., 2024). Ren & Sutherland (2025) describe a "squeezing" effect of DPO's negative gradient that pushes probability mass toward the model's top choice; our token-level probes did not find that signature here, and at matched SFT margin the rejected response's likelihood had no effect (Appendix A). Chen et al. (2024) show that DPO rarely changes the reference model's own preference rankings, which fits DPO spending most of its margin growth on pairs the reference already gets right. Per-pair weighting and margin-aware variants already exist (e.g. ODPO, Amini et al., 2024; β-DPO, Wu et al., 2024; WPO, Zhou et al., 2024), as do DPO losses with an added chosen-likelihood term (e.g. DPOP; Pal et al., 2024).

What this project adds:

- **A located cause in a fully open pipeline.** Stage-by-stage checkpoints and released data let the effect be pinned to one stage, then reproduced and manipulated by rerunning that stage.
- **A mechanism.** Among attention heads, DPO's extra confidence comes from what SFT's own answer heads write at the answer position (30 such heads carry 35% of it, 30 random heads 1%), while its MLP part is spread across layers; causal tests show it is not a uniform scale-up of those heads.
- **A data-level causal account.** The evidence points to DPO training on pairs SFT already gets right, from agreement-controlled data splits, preference-label swaps and how held-out margins change (Appendix A), with the key contrasts replicated in a pre-registered three-seed test.
- **An intervention that tests the account, and its limit.** Weighting pairs by the reference model's own agreement removes the effect at 25% scale and about 60% of it at full scale; the reference-free SimPO objective (Meng et al., 2024) removes it in small tests. A pre-registered test shows the reference is what matters: SimPO stays at SFT's calibration, and adding DPO's reference to it restores all of plain DPO's overconfidence.

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Appendix A: Earlier single-run tests and other evidence

These small tests at 8× Ai2's learning rate came first. Each is a single run; repeating a run on the same pairs moves T* by about ±0.02, while different random subsets differ by about 0.2. They add details the replication does not cover:

- **Agreement, not how unlikely the rejected response is.** At equal rejected-response probability under SFT, training sets with 91% vs 27% agreed pairs give T* 3.30 vs 1.80. At equal agreement, rejected responses of very different probability give the same T* (2.37 in both runs). These splits were not matched on data source or task type.
- **Swapping labels removes it.** Swapping chosen and rejected turns agreed pairs into disagreed ones. Normal pairs give T* 2.39, half of them swapped 1.50 (about SFT's 1.53), all of them swapped 1.12. The half-swapped run learns little, since its two halves pull in opposite directions; the all-swapped run clearly moved (its accuracy fell from .61 to .59).
- **Agreed margins get multiplied.** After plain DPO, a held-out pair's margin is roughly its SFT margin times a constant (the margin slope). For agreed pairs that constant is 1.25–1.48, so margins SFT already had grow in proportion to their size; for disagreed pairs it is 0.85–1.01, so they mostly shift instead. The multiplying is strongest for long responses, and long responses also sharpen more on their own: training sets of long vs short responses give T* 2.89 vs 1.92 at the same agreement (62–64%), for reasons we have not tested.

How much overconfidence results also depends on how far training moves the weights. Olmo 3's Think DPO pairs are even more agreed (91% under Think-SFT), yet Think DPO sharpens much less. Two candidate reasons, both untested: its 12.5× lower learning rate, and its responses reason before answering, which the Where section shows reduces the effect on MMLU.

The account also explains why Olmo 3's "delta-learning" pairs (Geng et al., 2025; chosen response from a stronger model, rejected from a weaker one) sharpen more than the pairs Ai2 ranked with an LLM judge when building the data. They are 74% agreed vs 54%, and a 25%-scale run on delta-learning pairs alone reaches T* 2.02, vs 1.85 on the mixed data.

Appendix B: Full-scale details

BigBenchHard (Suzgun et al., 2023) follows Ai2's published settings (Team Olmo, 2025; Gu et al., 2025): zero-shot, "Let's think step by step", sampled, no practical length limit, Ai2's answer extraction. It reproduces the direction of Ai2's SFT → DPO gain (51.0 → 69.3 in their report; .448 → .567 here) but not its size, since their exact configuration is not published. Our first harness (3-shot chain-of-thought, capped at 1,536 tokens) showed no DPO gain: with three worked examples SFT already scores .64, and the cap cut off under 4% of SFT's and DPO's answers (37% of the full-scale weighted model's).

Two explanations for the full-scale drift do not fit the data. The weights are computed once from SFT and go stale as the model learns, but the 25%-scale run, which held, was just as stale: in both, about half the gradient weight lands on pairs the current model already gets right (vs about a third at the start). And the weights do not favor long chosen responses. SFT gives longer responses higher per-token likelihood, so agreed pairs tend to be the ones whose chosen response is longer, and down-weighting them halves the data's preference for longer answers. What differs is how far training moves the model: the full run widens training-pair margins by +0.80 per token, four times the 25% run's +0.21 and more than released DPO's +0.65. Two readings, both untested: weighting lowers the sharpening per unit of movement, so at full scale's movement some remains; and because the rescaling nearly doubles the weight on disagreed pairs, the weighted run may simply move further than plain DPO.

Appendix C: Mechanism details

- **About 2/3 signal, 1/3 normalization.** Of the ×1.44, ×1.27 comes from larger head and MLP outputs along the letter directions and ×1.14 from a smaller final residual norm, which the scores are divided by. We ruled out one proposed normalization mechanism: last-layer "entropy neurons" (Stolfo et al., 2024) do change with DPO, but swapping their activations between SFT and DPO has no effect.
- **More sharpening, larger multiplier.** ×1.20 for our 25%-scale rerun (T* 1.85), ×1.44 for both a small test (T* 2.40) and the released model (T* 2.34), with largely the same top heads.
- **Agreement weighting and SimPO remove it, up to 25% scale.** Their letter scale is ×0.99–1.03 relative to SFT in small tests and ×0.99 at 25% scale, and agreed held-out margins are no longer multiplied (margin slope 0.90–0.93 vs 1.35 for plain DPO in small tests; 0.97 vs 1.27 at 25% scale).

- **At full scale part of it returns, by another route.** The full-scale weighted model's letter scale is ×1.12 (released DPO ×1.44), led by the same top head (L22H6 +0.14 vs DPO's +0.26). But its answer signal grows only ×1.04 (DPO ×1.27), while the final residual norm shrinks almost as in DPO (×1.08 vs ×1.14). The weighting removes about 80% of DPO's signal gain but only about 40% of its normalization change, so most of the remaining overconfidence runs through the normalization route. The shrinkage is in directions the output layer barely reads: their share of the final residual drops from .62 to .57 in both released DPO and the full-scale weighted model, but not at 25% scale (.62).

For free-form answers, the top-two gap at the first token of short answers grows ×1.44, but through different, later heads (L27H7, L31H23). So the effect is a general change in how firmly the model commits to an answer.